

Effects of Diameter on the Durability of Glass Fiber-Reinforced Polymer Bars Conditioned in Alkaline Solution

Brahim Benmokrane¹; Allan Manalo²; Jean-Charles Bouhet³; Khaled Mohamed⁴; and Mathieu Robert⁵

Abstract: Current standards do not consider the diameter of glass fiber-reinforced polymer (GFRP) bars used as internal reinforcement in concrete structures to be a factor influencing bar durability. This paper investigates the effects of bar diameter on the physical and mechanical properties as well as the durability of GFRP reinforcing bars conditioned for three months at 60°C in an alkaline solution simulating a concrete environment. Five diameters (nominal diameters of 9.5, 12.7, 15.9, 19.1, and 25.4 mm) were considered; bar properties were assessed before and after conditioning. Microstructural analyses and measurement of physicochemical properties were also carried out. The results show that bar size had no significant effect on bar physical properties, except for water absorption. The smaller diameter bars had higher water absorption than the larger ones because of their higher surface area-to-volume ratios. In the case of the unconditioned bars, the tensile strength and modulus were not significantly affected by bar diameter, but there was a size effect for interlaminar shear strength and flexural strength. Conversely, the conditioning in the alkaline solution had a greater negative effect on the tensile strength of the larger bars than on the smaller ones. Scanning electron microscope (SEM) observations and Fourier transform infrared spectroscopy (FTIR) analysis revealed that the degradation remained at the surface of all the conditioned specimens. Nevertheless, there were only small variations between the physical and mechanical properties of the GFRP bars of different diameters. This indicates that the current provisions in standards that do not relate strength retention limit to bar size are acceptable. DOI: 10.1061/(ASCE)CC.1943-5614.0000814. © 2017 American Society of Civil Engineers.

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Introduction

Glass fiber-reinforced polymer (GFRP) bars have emerged as an attractive alternative to steel reinforcement in concrete structures because of their corrosion performance in aggressive environmental conditions. These composite reinforcing bars have attracted significant interest because of their superior properties such as light weight, high mechanical properties, and neutrality with respect to electrical and magnetic disturbances. The results of several experimental studies, establishment of materials specifications,

publication of design codes and guidelines, and successful field applications in concrete structures have driven the worldwide use and acceptance of GFRP bars [ACI 440.1R (ACI 2015); CSA S6 (CSA 2006), CSA S806 (CSA 2012), CSA S807 (CSA 2010); *fib* 2007; Benmokrane et al. 2006; Drouin et al. 2011; Manalo et al. 2014; Mohamed and Benmokrane 2014, 2015; Arafa et al. 2016; Ahmed et al. 2016]. As a result, a variety of GFRP bars manufactured with various fibers and resins as well as various surface geometries are now commercially available [ACI 440.1R (ACI 2015)]. Like steel reinforcement, GFRP bars are available in diameters ranging from 6 to 36 mm [ACI 440.1R (ACI 2015)]; larger diameter bars can be also manufactured. Unlike steel reinforcement, whose properties can be assumed to be the same for different bar diameters, GFRP bars are size-dependent in terms of longitudinal strength due to shear-lag effect (Bank 2006). To illustrate, Hollaway (2008) measured a reduction in tensile strength of up to 40% when the bar diameter increased from 9.5 to 20 mm. For these reasons, manufacturers are required to fully report the characteristic strength, stiffness, physical, and durability properties for every type and size of fiber-reinforced polymer (FRP) bar.

The mechanical, physical, and durability characteristics are important information needed in the material specifications and design of FRP bars. These properties should be determined in accordance with the prescribed test standards and methods [ACI 440.6 (ACI 2008); CSA S807 (CSA 2010)]. They must also be made available to design engineers, asset owners, and building-code and standard-writing authorities (Micelli and Nanni 2004). More importantly, the physical, mechanical, and durability properties of every commercially available and newly developed GFRP bar should comply with the limits specified in the current standards

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and specifications [ACI 440.6 (ACI 2008); CSA S807 (CSA 2010)]. While the minimum specified mechanical properties are available for each bar diameter, the current ACI and CSA standards and specifications [ACI 440.6 (ACI 2008); CSA S807 (CSA 2010)] do not relate the requirement limits in terms of physical and durability properties to FRP bar size. This is because there have been few studies investigating the effect of GFRP bar size on physical and durability properties.

The durability of GFRP bars is a complex problem because it depends on the components of the composite material (Bakis 1993; Benmokrane and Rahman 1998; Hollaway 2010). In particular, the reaction of GFRP bars to the alkali in concrete has received significant attention because of their importance in construction applications (Porter et al. 1997; Bakis et al. 1998; Nkurunziza et al. 2005; Chen et al. 2007; Robert and Benmokrane 2013; Belarbi and Wang 2012; Kamal and Boulfiza 2011). As internal reinforcement, FRP bars are embedded in a cementitious environment aggressive to GFRPs because of the high pH level of the pore-water solutions and the presence of alkali ions (Porter et al. 1997; Micelli and Nanni 2004; Robert et al. 2009). Most of the cases available to date focusing on the physical and durability properties of GFRP bars have involved small-diameter bars. Tannous and Saadatmanesh (1999) investigated the durability characteristics of 10 and 19.5 mm diameter vinylester and polyester-based AR-Glass FRP bars. Their results showed very small changes in the elastic modulus (0–2%). The tensile strength retained after six months exposure to an alkaline solution (pH = 12) at 60°C was only 72–77% for 10 mm bars, but 83–86% for 19.5 mm diameter bars. Furthermore, they indicated that vinylester provided the fibers with better protection against chemical attack than did polyester. Similarly, Micelli and Nanni (2004) investigated the durability of GFRP rods subjected to alkaline exposure and high temperature (60°C). While they considered two different diameters of GFRP rods, the 12 mm diameter bars were made from thermoplastic resin, while the 6.35 mm diameter bars were made from polyester resin. Their results revealed no degradation in the GFRP bars made with thermoplastic resin, but a reduction in tensile strength of up to 40% in the bars made with polyester resin after 42 days. Robert et al. (2009) conducted an accelerated aging test of 12.7 mm vinylester-based GFRP bars with a fiber content of 77.9% by weight embedded in concrete. They found that the GFRP bars in alkaline solution experienced strength losses more than three times that of the GFRP bars aged in moist concrete. Most recently, Benmokrane et al. (2015) conducted a comparative durability study of 6 mm glass/vinylester, basalt/vinylester, and basalt/epoxy FRP bars. Their test results revealed that the basalt/epoxy FRP bars yielded better retention of mechanical properties than did the basalt/vinylester ones. The results from these studies primarily provide the basis that the current design guidelines and codes can use in developing generalized requirement limits as well as similar environmental reduction factors for all GFRP bar sizes. Clearly, there is a gap in research on the durability characteristics of GFRP bars of different sizes and on correlating these characteristics with important physical and mechanical properties.

This paper systematically investigates the physical and mechanical properties of GFRP bars of different diameters. It aims to correlate the physical and long-term characteristics of FRP bars with bar diameters for their effective use and to provide guidance in the development of design codes and material specifications for this reinforcing material. In the first stage, the physical and chemical properties of the unconditioned bar materials were determined. In the second stage, the GFRP bars were exposed to alkaline solution for three months at 60°C to simulate the concrete pore environment. Mechanical characterization of these conditioned bars was then

conducted and compared to that of the unconditioned bars. Microstructural analyses and measurements of the physicochemical properties were also carried out on the conditioned and unconditioned GFRP bars. The findings from these studies are provided in this paper.

Experimental Program

Materials

The sand-coated GFRP bars used in this study were made of continuous boron-free glass fibers (EC-R) impregnated in a vinylester-based resin matrix and were manufactured according to the pultrusion process by a Canadian company (Pultrall, Thetford Mines, Quebec, Canada). Five diameters of GFRP bars were investigated (#3, #4, #5, #6, and #8), which correspond to nominal diameters of 9.5, 12.7, 15.9, 19.1, and 25.4 mm, respectively, as shown in Fig. 1. The technical specifications for these bars can be found in the manufacturer's data sheets (Pultrall 2016).

Specimen Details

The preparation of specimens and characterization of the physical and mechanical properties of the GFRP bars were performed according to the appropriate ASTM, American Concrete Institute (ACI), and Canadian Standards Association (CSA) test standards. Table 1 summarizes the test methods as well as the number of specimens tested for each type of test and bar size. The specimens were cut and prepared in accordance with the recommendation of the appropriate test standards.

Bar Conditioning

The GFRP bars were separated into two series. The first series consisted of unconditioned reference bars; the second comprised GFRP bars conditioned in an alkaline solution for 90 days at 60°C. The alkaline solution used comprised 118.5 g of $\text{Ca}(\text{OH})_2$, 0.9 g of NaOH, and 4.2 g of KOH for 1 L of water. The solution had a pH of 12.6, which is representative of a mature concrete pore solution. The conditioning temperature was set at 60°C, as specified in ASTM D7705/D7705M-12 (ASTM 2012). The conditioning was conducted in accordance with ACI 440.3R-12 (ACI 2012), Test method B.6, and CSA-S806-12 (CSA 2012), Annex O. During conditioning, the level of alkaline solution and pH level were checked periodically, and new solution added as necessary.

Physical Properties of the GFRP Bars

Table 2 summarizes the physical properties of the GFRP bars. The values listed within parentheses represent the standard variation of the test results.



Fig. 1. Tested GFRP bars specimens

Table 1. Summary of the Test Methods and Number of Specimens

Properties	Test method	Number of specimens	Reference	Conditioned
Physical properties				
Cross-sectional area (mm ²)	CSA-S806, Annex A (CSA 2012)	9	Yes	—
Fiber content by weight (%)	ASTM D3171-15 (ASTM 2015)	9	Yes	—
Transverse CTE ($\times 10^{-6}/^{\circ}\text{C}$)	ASTM E1131-08 (ASTM 2014b)	9	Yes	—
Void content (%)	ASTM D5117-09 (ASTM 2009)	9	Yes	—
Water absorption at 24 h (%)	ASTM D570-98 (ASTM 2010)	15	Yes	—
Water absorption at saturation (%)	ASTM D570-98 (ASTM 2010)	15	Yes	—
Cure ratio (%)	ASTM E 1356-08 (ASTM 2014a)	15	Yes	—
T_g ($^{\circ}\text{C}$)	ASTM E 1356-08 (ASTM 2014a)	15	Yes	—
Mechanical properties				
Flexure	ASTM D4476/D4476M-14 (ASTM 2014c)	6	Yes	Yes
Interlaminar shear	ASTM D4475-02 (ASTM 2016)	6	Yes	Yes
Tensile	ASTM D7205/D7205M-06 (ASTM 2011b)	6	Yes	Yes

Table 2. Physical Properties of the GFRP Bars

Property	Bar designation				
	#3	#4	#5	#6	#8
Nominal diameter (mm)	9.5	12.7	15.9	19.1	25.4
Nominal cross-sectional area (mm ²)	70.8	126.6	198.5	286.5	506.7
Actual cross-sectional area by immersion test (mm ²)	83.8 (1.9)	145.0 (1.7)	224.4 (1.2)	317.3 (1.9)	555.7 (5.2)
Fiber content by weight (%)	80.9 (0.2)	81.8 (0.1)	82.6 (0.1)	82.7 (0.2)	83.0 (0.2)
Transverse CTE ($\times 10^{-6}/^{\circ}\text{C}$)	20.7 (2.3)	20.5 (1.6)	21.5 (1.4)	22.0 (1.8)	21.3 (2.5)
Void content (%)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Water absorption at 24 h (%)	0.15 (0.01)	0.12 (0.02)	0.10 (0.01)	0.02 (0.00)	0.02 (0.00)
Water absorption at saturation (%)	0.19 (0.01)	0.15 (0.05)	0.14 (0.01)	0.04 (0.01)	0.04 (0.00)
Cure ratio (%)	100 (0)	100 (0)	100 (0)	100 (0)	100 (0)
T_g ($^{\circ}\text{C}$)	125.8 (1.3)	112.8 (2.5)	109.6 (1.5)	105.2 (1.3)	124.8 (4.9)

Actual Cross-Sectional Area by Immersion Test

The actual cross-sectional area of the GFRP bars was measured in accordance with CSA-S806 (CSA 2012), Annex A. A plastic cylinder and a scale capable of measuring weight up to an accuracy of 1 g were used. Twenty-four 270 mm long specimens were prepared for each bar diameter and tested. All specimens were kept in the test environment for 24 h prior to weighing and measuring. All the bars are oversized when compared to their respective nominal cross-sectional areas (Table 2). The average measured oversize varied from 9 to 18%, as seen with the largest (#8) and smallest (#3) bars, respectively. This variation is due to the ratio difference of the sand coating to the core. According to the average measured surface area, all bar diameters meet the Grade III requirements of the Ministry of Transportation Ontario's special provision for glass fiber-reinforced polymer reinforcing bar (MTO 2014).

Fiber Content

The fiber content of the GFRP bars was calculated according to ASTM D3171-15 (ASTM 2015). Nine specimens for each bar diameter were identified, dried, and weighted. The bars were then placed in an oven at 650°C until the polymer matrix was entirely removed by combustion. The remaining fibers were weighed in order to get the fiber weight ratio. Because the bars were sand coated, the weight of the sand was measured separately and subtracted from the initial weight. Table 2 shows that the fiber content increased slightly with bar diameter, from 80.9% of fibers by weight for the #3 bars to 83.0% for the #8 bars. However, that the limits specified in CSA S807 (CSA 2010) for the glass fiber fraction by weight is only 70%.

Transverse Coefficient of Thermal Expansion

The transverse coefficient of thermal expansion was calculated according to ASTM E1131-08 (ASTM 2014b). Nine specimens were tested for each bar diameter. The measurements were conducted between -30 and 60°C at a heating rate of 3°C . A TA Q400 thermomechanical analyzer was used. Cryogenic equipment (liquid nitrogen) was used to reach subzero temperatures. The results show that the coefficients of thermal expansion for the different bar diameters fall between $20.5 \times 10^{-6}/^{\circ}\text{C}$ and $22.0 \times 10^{-6}/^{\circ}\text{C}$, which is only half of the limit of $40.0 \times 10^{-6}/^{\circ}\text{C}$ specified in CSA 807 (CSA 2010).

Void Content

The void content of the GFRP bars was measured with the wicking test according to ASTM D5117-09 (ASTM 2009). Five specimens 25 mm in length for each bar diameter were placed into a dye penetrant solution from a mixture of basic fuchsin ($\text{C}_{20}\text{H}_{19}\text{N}_3\text{HCl}$) and methyl hydrate with a 1:100 weight ratio. The wicking action is revealed by the apparition of spots on the side of the specimen not immersed in the solution. The tests were made at a room temperature and relative humidity of 23°C and 50%, respectively. Before the test, a precautionary nail-polish shield recommended by ASTM D2374-05 (ASTM 2011a) was applied in order to prevent wicking along the bar periphery. The dye penetrant solution is poured in a disposable pan so that only the bottom 3 mm of the specimens is immersed. The presence and number of the colored dots were then noted after 15 min of immersion. No colored dots were observed in any of the specimens, indicating that there were no voids or cracks running through the length of the bars. As

specified in CSA 807 (CSA 2010), the void content in FRP bars should be no greater than 1%.

Water Absorption

The water absorption of the bars after 24 h and at saturation was determined according to ASTM D 570-98 (ASTM 2010). Nine specimens 75 mm in length were prepared, dried, and weighed for each bar diameter. These specimens were then entirely immersed in distilled water at 60°C. The samples were removed from the water after 24 h, surface dried, and weighed. Then, they were placed in water again until full saturation, i.e., when the weight increase in three consecutive weightings was less than 1%. In calculating the water absorption, the loss of sand coating was considered by weighing the specimens two separate times, i.e., just after removing the specimens from water, and after 24 h of oven drying at 60°C. The difference between the two measurements gave the real mass of absorbed water.

The water absorption of the GFRP bars at 24 h and at saturation was found to decrease as the diameter of the bars increased. The maximum absorption of the #3 bars after 24 h and at saturation was 0.152 and 0.195%, respectively. This low water absorption for the GFRP bars considered in this study—even at saturation—is due to the low moisture diffusion of the vinylester resin, as indicated by Tannous and Saadatmanesh (1999). For FRP reinforcing bars, the limit of water absorption after 24 h specified in CSA-S807 (CSA 2010) is 0.25%. The values at saturation are 1 and 0.75% (high durability) in ACI 440.6M (ACI 2008) and CSA-S807 (CSA 2010), respectively. This shows that the water absorption of the GFRP bars with different diameters is well within the allowable specified limit in current standards.

Cure Ratio and Glass Transition Temperature

The cure ratio and glass transition temperature, T_g , of the GFRP bars were determined by differential scanning calorimetry (DSC) according to the ASTM E 1356-08 (ASTM 2014a) test method. For this test, nine pieces weighing approximately 20 mg were cut from the center of the core of each bar, weighed, and placed in an aluminum pan. The specimens were then heated from 25 to 180°C at a rate of 10°C/min. The T_g obtained for all bar diameters ranged from 105 to 125°C for the GFRP bar, which is higher than the limit of 100°C specified in CSA S807 (CSA 2010) and ACI 440.6M (ACI 2008). Similarly, a cure ratio of 100% was measured for all bar diameters. It is worth noting that

the specified cure ratio for GFRP bars is only 95% [CSA S807 (CSA 2010)].

Mechanical Properties of the GFRP Bars

The mechanical properties of the unconditioned (reference) and conditioned GFRP bars with different diameters were assessed under three-point flexural testing, short-beam shear testing, and tensile testing. Because of the limited length of both the conditioned and unconditioned #6 GFRP bar specimens provided by the industry partner, there were no portions remaining to conduct the flexural testing for this bar size. Table 3 summarizes the results of the mechanical characterization. The values listed within parentheses are the standard deviation of the test results. The nominal diameter and nominal cross-sectional area of the GFRP bars were used in calculating the mechanical properties.

Flexural Properties

Flexural testing was conducted for comparative assessment of the mechanical properties of the FRP bars. The test was conducted in accordance with ASTM D4476/D4476M-14 (ASTM 2014c). The unconditioned and conditioned specimens were tested over a simply supported span equal to 20 times the bar diameter and with an overhang twice the bar diameter at each support. Six replicates for each bar diameter were tested under laboratory conditions on a Material Testing Systems MTS 810 testing machine equipped with a 500 kN load cell. The specimens were loaded at the midspan with a circular nose and supported at the ends on two circular supports that allowed the specimens to bend, as shown in Fig. 2(a). A displacement control rate of 3.0 mm/min was used. The applied load and deflection were recorded during the test with a data acquisition system monitored by a computer. The flexural strength of the GFRP bars, f_u , in the outermost fibers at midspan was calculated as $f_u = PLc/(4I)$, where P is the failure load; L is the clear span; c is the distance to the centroid of the extreme-most fibers; and I is the moment of inertia.

Under bending, the load-deflection behavior of all the specimens [Fig. 3(a)] showed linear behavior, but a slight reduction in stiffness was observed before final failure. This reduction of stiffness is attributable to the initiation of compressive failure under the loading point, which is more noticeable in the conditioned specimens than the reference samples. Moreover, the conditioned samples failed at lower load and deflection than the reference samples, but with the same bending stiffness prior to failure. Regardless of diameter, the bars failed because of compression in the top fibers,

Table 3. Mechanical Properties of the GFRP Bars before and after Conditioning

Property	Bar designation				
	#3	#4	#5	#6	#8
Reference					
Flexural strength, f_u (MPa)	1,623.7 (58.2)	1,588.1 (93.5)	1,757.5 (66.5)	—	1,406.3 (151.3)
Interlaminar shear strength, S_u (MPa)	54.7 (1.1)	52.9 (2.1)	55.8 (1.5)	56.0 (0.1)	47.5 (3.7)
Tensile strength, f_t (MPa)	1,315.3 (31.1)	1,281.5 (35.3)	1,237.4 (33.3)	1,270.0 (31.4)	1,271.0 (29.9)
Tensile modulus, E (GPa)	62.5 (0.4)	61.3 (0.4)	60.0 (1.3)	60.5 (0.5)	61.8 (0.3)
Tensile strain, ϵ	2.3 (0.1)	2.1 (0.1)	2.1 (0.1)	2.1 (0.1)	2.1 (0.1)
Conditioned					
Flexural strength, f_u (MPa)	1,440.3 (97.3)	1,455.6 (96.4)	1,660.0 (62.3)	—	1,370.9 (49.9)
Interlaminar shear strength, S_u (MPa)	48.2 (1.5)	50.6 (2.1)	50.2 (2.4)	52.1 (1.8)	47.9 (2.4)
Tensile strength, f_t (MPa)	1,251.8 (34.6)	1,114.0 (29.4)	1,141.1 (41.5)	1,095.9 (55.4)	1,119.2 (50.0)
Tensile modulus, E (GPa)	63.0 (0.4)	61.9 (0.4)	60.1 (0.2)	60.8 (0.6)	61.7 (0.3)
Tensile strain, ϵ	2.2 (0.1)	1.8 (0.1)	1.9 (0.1)	1.8 (0.1)	1.8 (0.1)

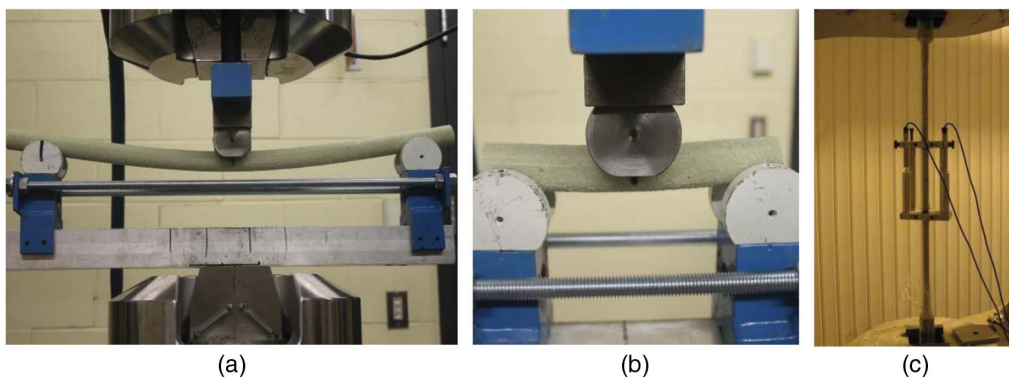


Fig. 2. Test setup and instrumentation: (a) flexure; (b) short-beam shear; (c) tensile

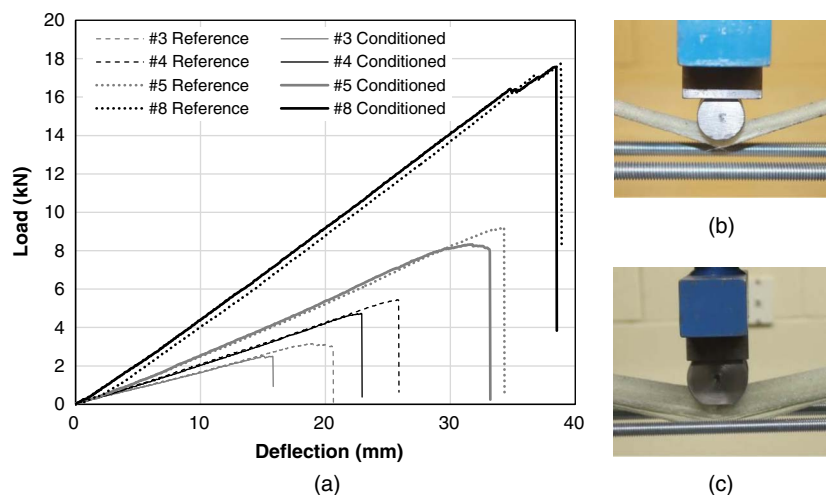


Fig. 3. Load-deflection and failure behavior of GFRP bars in bending: (a) load and deflection behavior; (b) failure of #3 bars; (c) failure of #8 bars

followed by tensile failure at the bottom near the midpoint of the specimens [Figs. 3(b and c)].

Interlaminar Shear-Strength Properties

The short-beam shear testing was conducted following ASTM D4475-02 (ASTM 2016) in order to determine the interlaminar shear strength of FRP bars [Fig. 2(b)]. The short-beam shear test is a matrix-dominated property and can give an indication of the resistance of the fiber-matrix interface. The test was carried out with Material Testing Systems MTS 810 testing machine equipped with a 500 kN load cell. Six replicates for each bar diameter were prepared and tested. The distance between the shear planes was set to six times the nominal diameter of the FRP bars. A displacement control rate of 1.3 mm/min was employed for the #3, #4, and #5 bars, while 1.8 mm/min was used for the #6 and #8 bars. The applied load and displacement were recorded with a computer-monitored data-acquisition system. The interlaminar shear strength S_u of the FRP bar was calculated as $S_u = 0.849P/d^2$, where P is the shear failure load; and d is the bar diameter.

Fig. 4 shows the results from short-beam shear testing. It is shown that the applied load increased linearly with the deflection with a slight nonlinearity before the final failure. This nonlinear behavior before final failure is more apparent in the larger diameter

bars. The experimental results also show that the conditioned specimens failed at a lower load than the reference samples. Micelli and Nanni (2004) indicated that the decrease in the apparent horizontal shear strength of the conditioned GFRP bars is caused by resin damage attributable to penetration of the alkaline solution. Both the unconditioned and conditioned specimens failed because of horizontal shear cracks that originated from the edge of the bars and developed along the length [Figs. 4(b and c)]. Park et al. (2008) indicated that very high interlaminar-shear stresses can arise at the free edge of fiber composite materials. The only observed difference was the more obvious compressive failure under the loading point and at the supports that occurred with the larger diameter bars. This is because of the higher load needed to cause failure of the larger diameter bars and achieve the same level of interlaminar shear stress, while the contact area under the loading point was almost the same for all of the bar sizes. This observation explains the nonlinear behavior of the bars before the final failure.

Tensile Properties

Tensile testing was conducted according to ASTM D7205/D7205M-06 (ASTM 2011b) and CSA-S806-12 (CSA 2012) Annex C. The tensile testing along the alignment of fibers is related to fiber properties. The gauge length of the specimens was

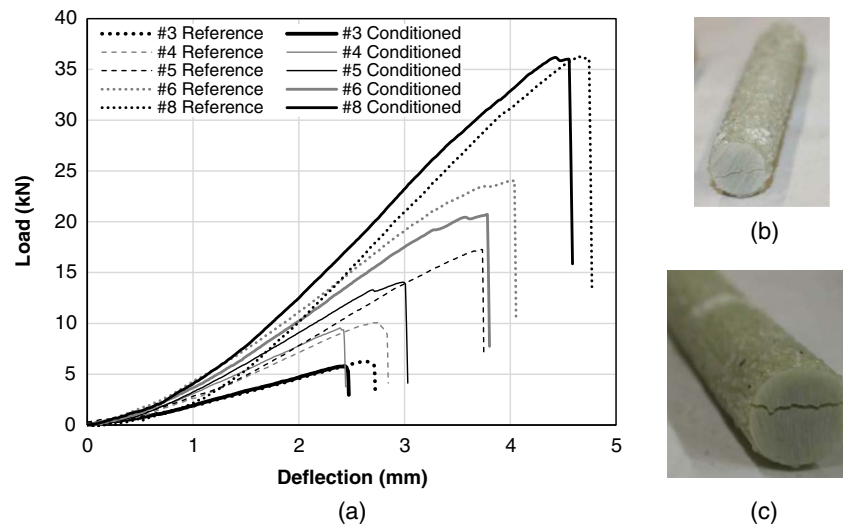


Fig. 4. Load-deflection and failure behavior of the GFRP bars under short-beam shear testing: (a) load and deflection behavior; (b) failure of #3 bars; (c) failure of #8 bars

Table 4. Specimen Length for the Tensile Testing of the GFRP Bars

Bar size (#)	Gauge length (mm)	Anchor length (mm)	Specimen length (mm)
3	465	400	1,265
4	583	510	1,603
5	713	630	1,973
6	848	740	2,328
8	1,166	990	3,146

approximately equal to 40 times the bar diameter, in addition to the length of the anchorage steel tubes at each end of the GFRP bars, as specified in CSA-S806-12 (CSA 2012), Annex B. Table 4 provides the specimen and anchor length for each bar size. Each specimen was instrumented with two LVDTs 200 mm in length to capture specimen elongation during testing [Fig. 1(c)]. To avoid damaging the LVDTs, they were detached from the specimen when the load reached 75% of the estimated ultimate load. The tests were carried out with a Satec-Baldwin testing machine equipped with a 2,000 kN load cell. The load was increased at a rate of

300 MPa/min until tensile failure occurred. Six specimens were tested for each bar diameter. The applied load and bar elongation were electronically recorded during the test with a computerized data-acquisition system. This test determined the ultimate tensile strength f_t , tensile modulus E , and tensile strain ϵ .

The typical stress strain behavior of the GFRP bars with different diameters is shown in Fig. 5(a). All of the specimens behaved in a linear elastic fashion in tension up to failure and exhibited almost identical slopes in the stress-strain curve. Note that the strain after the LVDTs were removed was calculated based on the stress and elastic moduli of the bars. This behavior indicates that the elastic moduli of the bars with different diameters were similar and that the loss of elastic modulus due to exposure to the simulated alkaline environment was negligible. Nevertheless, the failure stress and rupture strain of the conditioned bars were lower than that of the reference bars. Robert et al. (2009) had similar findings and they indicated that the elastic modulus of the GFRP bars they tested was not affected by aging in a concrete environment, but tended to be more brittle and evidenced lower strength than the reference bars. Generally, rupture strain decreases as bar diameter increases. Figs. 5(b and c) show that, regardless of diameter, the GFRP bars

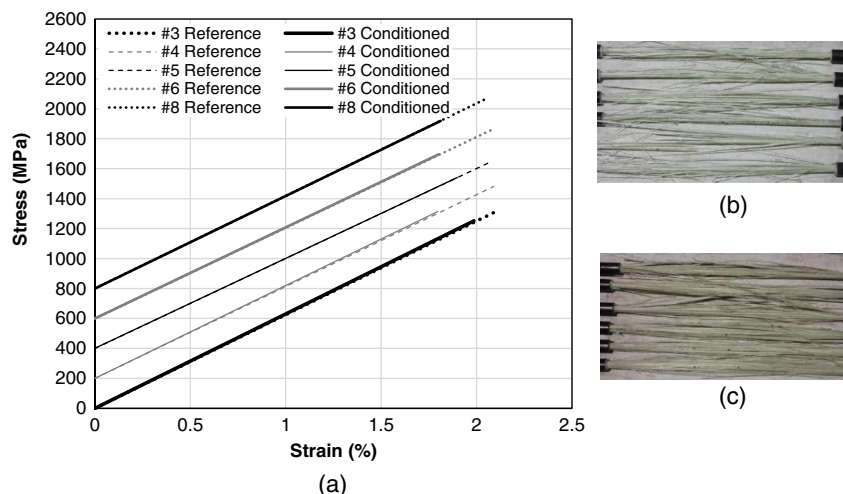


Fig. 5. Stress-strain and failure behavior of the GFRP bars in tension: (a) stress and strain behavior; (b) failure of #3 bars; (c) failure of #8 bars

tested failed at the middle of the bar (within the gauge length). As expected, all of the specimens failed suddenly due to tensile fiber rupture. Prior to failure, a popping noise was heard caused by some of the fibers and/or the resin failing on the outer perimeter of the bar. The measured tensile strength and elastic modulus for both the reference and conditioned bars (all bar diameters) was significantly higher than the minimum specified mechanical properties in ACI 440.6M (ACI 2008) and CSA S807-10 (CSA 2010) for high modulus GFRP bars. Similarly, the failure strain was higher than the prescribed 1.2%.

Discussion

The effects of bar diameter on physical, mechanical, and durability properties are analyzed and discussed in this section.

Effect of Bar Diameter on Physical Properties

The bar diameter had no significant effect on most of the physical properties of the GFRP bars, including the transverse coefficient of thermal expansion, porosity, and T_g . Similarly, all of the bars tested evidenced an entirely cured resin, indicating that bar diameter did not affect the degree of cure. The development of an efficient production method makes this consistency possible. This is in contrast with the observations made by Yi and Hilton (1998), who indicated that laminate thickness might affect the degree of cure due to the higher thermal conductivity of thicker composite laminates. On the other, the fiber content and water absorption were found to increase and decrease, respectively, with increasing bar diameter. Because none of the FRP bars contained voids, the lower water absorption for the larger diameter bars can be correlated to increasing fiber content (by weight), as noted in Table 2. Glass fibers absorb scarcely any water, therefore the bars with higher matrix contents evidenced higher absorption rates.

In order to further correlate bar diameter to the percentage water absorption, the shape ratio of the GFRP bars were calculated and plotted (Fig. 6). Cinquin and Medda (2009) defined the shape ratio as the ratio between the sample's surface and volume. As shown in Fig. 6(a), the shape ratio was significant in the water absorption of the GFRP bars at 24 h and at saturation, i.e., the water absorption increased as did the shape ratio. It can also be observed that the relationship between the water absorption at 24 h (%) and at saturation to that of the shape ratio is the same, as demonstrated by the almost equal slopes of the water absorption and shape-ratio

relationship curves. Conversely, there is a linear but negative correlation between the shape ratio and bar diameter. The shape ratio decreases as the bar diameter increases. This accounts for the smaller diameter bars having higher absorption rates than the larger diameter ones with the same length because the exposed surface is greater with respect to volume. Note that the decrease in the shape ratio is very similar to the decrease in the percentage of water absorption at saturation for the various bar diameters.

Effect of Bar Diameter on Mechanical Properties

Many studies have revealed that the short-term mechanical properties of FRP bars decrease with increased bar diameter (Bank 2006; Holloway 2008). However, this conclusion was not clearly observed in this study. Fig. 7 shows the normalized mechanical properties for the different sizes of GFRP bars. This graph provides the percentage of the interlaminar shear strength (ILSS), flexural strength (flexure), tensile strength (tensile), and tensile modulus (modulus) for all of the bar diameters with respect to the mechanical properties of the #3 bars. The figure show no significant difference in the tensile properties of the GFRP bars regardless of bar diameter. While the highest tensile strength and modulus were observed for the #3 bars, the lowest tensile properties were exhibited by the #5 bars (94% compared with the #3 bars), with the #8 bars exhibiting more than 96% of strength and stiffness of the #3 bars. Kocaoz et al. (2005) suggested that modulus, which is an intrinsic property of the material, is not significantly affected by bar cross-sectional size but rather by the level of fibers contained in the bar. Because the percentage fiber content by weight (Table 2) for different bar diameters was almost the same, then the bars should record the same elastic modulus. Moreover, the consistency in the measured tensile properties of the GFRP bars with different diameters can be attributable to an adequate anchor length, which resulted in a more efficient transfer of stresses from the bar surface to the center. Portnov and Bakis (2008) suggested that introducing a more uniform distribution of the applied shear stress near the grips could minimize the shear lag effect and increase the tensile-load-carrying capacity of round pultruded rods.

In contrast to the tensile properties, there was a size effect observed for ILSS and flexural strength. The #8 bars recorded ILSS and flexural strength almost 14% lower than the #3 bars. Significant size effects were also observed by Wisnom and Jones (1996) on the average ILSS for unidirectional glass fiber/epoxy composites. They indicated that the lower ILSS with bigger specimens could be attributable to the larger inherent defects. Koller et al.

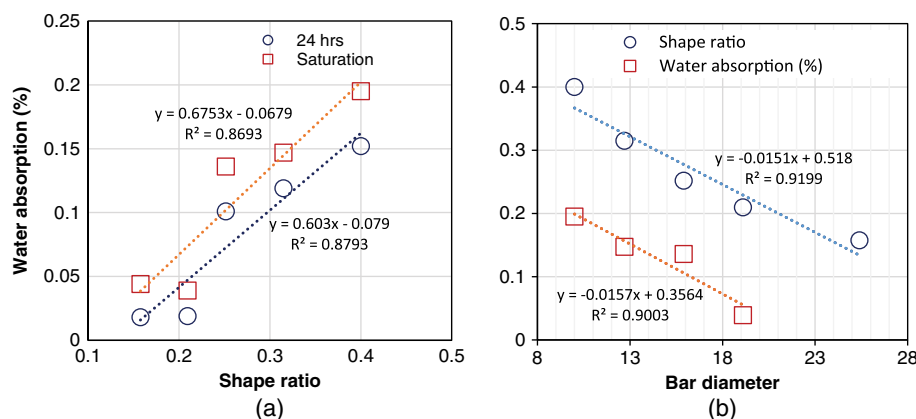


Fig. 6. Relationship of water absorption to the shape ratio and bar diameter: (a) water absorption and shape ratio; (b) shape ratio and water absorption with bar diameter

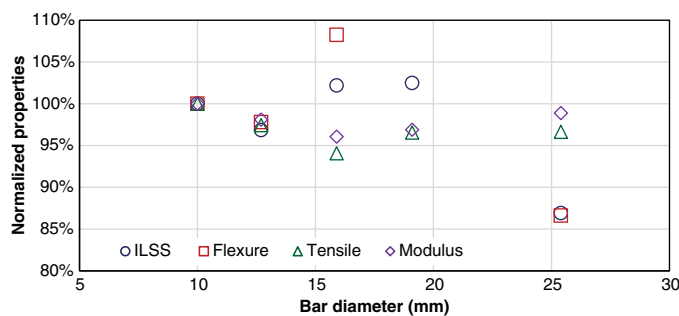


Fig. 7. Normalized mechanical properties of the GFRP bars

(2007) also suggested that the probability of having large defects in composite materials increases with increased specimen volume. Moreover, the ILSS is a matrix-dominated property. Wisnom and Jones (1996) suggested that matrix-dominated failures show the largest size effects in composite materials. Defects near the edge of the GFRP bars were very critical for the specimens subjected to the short-beam test as this location is subjected to higher levels of shear stress. Similarly, the lower flexural properties of the larger diameter bars can be explained by the high probability of defects. Carvelli et al. (2009) suggested that it is more difficult to keep the filaments parallel to one another in larger diameter FRP bars during the pultrusion process, increasing their tendency to buckle under compression. This is the failure mechanism observed in the GFRP bars during the flexural test, in which the failure was initiated by the compression buckling of the top fibers.

Property Retention of GFRP Bars with Different Diameters

Tannous and Saadatmanesh (1999) indicated that the effect of an alkaline solution on FRP bars only involved a very thin layer on the exposed surface. Thus, the approximate layer thickness and area of the GFRP bars affected by the alkaline solution were calculated and reported in Table 5 to correlate with the property retention for different bar diameters. These values were determined by assuming that the affected portions of the GFRP bar were the same as the percentage of water absorption at saturation and equally distributed along the surface of the bars (Table 2). This approach is similar to the method adopted by Cinquin and Medda (2009). It is shown that the affected thickness is only in the order of 1.9×10^{-3} to 5.4×10^{-3} mm. Moreover, the percentage of affected thickness with respect to the nominal diameter decreased as did bar diameter.

Fig. 8 provides the strength and stiffness retention properties of the GFRP bars with different diameters. Figs. 8(a–c) clearly indicate that the strength properties were affected by conditioning in the alkaline solution, while Fig. 8(d) shows that the modulus of elasticity was not affected. The almost 100% retention of the modulus of elasticity, which is a fiber-dominated property, for all bar diameters indicates that the damage caused by the moisture diffusion was confined to the very thin layer of the exposed surface and that the reinforcing fibers were not affected by the conditioning. As

the modulus of elasticity was determined from the linear-elastic portion of the stress and strain curve, the matrix was still effectively transferring stresses to the fibers. Nkurunziza et al. (2005) indicated that the diffusion of moisture and alkalis in composites can destroy the bond between the fiber and matrix, damaging the interface. This is difficult to see at lower loads, but higher mechanical loads increase the degradation of the fiber-matrix interface. This accounts for the noticeable decrease in strength properties, as these values were calculated at the point of failure of the GFRP bars at which the stress distribution along the fibers is less uniform.

Fig. 8(a) shows that the effect of conditioning on the ILSS of the GFRP bars was significant with the smaller diameter bars, but decreased as the diameter increased. In fact, the residual horizontal shear strength for the #8 bars was the same as that measured on the reference bars, while the #3 bars retained only 88% of their ILSS. This can be explained by the thinner layer affected by the alkaline solution in the case of the larger diameter bars, as reported in Table 5. As indicated in the previous section, ILSS is governed by the fiber-matrix interface. Thus, the decrease in ILSS can be correlated to interface degradation. Similarly, because of the very thin layer affected by conditioning in the alkaline solution in the larger diameter GFRP bars, its affected area was also significantly smaller with respect to the total area, compared with the smaller diameter bars.

Fig. 8(b) shows that the flexural-strength retention increased with bar diameter. The retention ranged from 88% retention for the #3 bars to more than 97% for the #8 bars. This behavior was also observed by Maranan et al. (2014), who found that larger diameter bars exhibited a slower rate of strength degradation at high temperature than small diameter bars. Cinquin and Medda (2009) also concluded that the residual mechanical properties were more affected in thin composites than thick composites. The higher flexural strength retention for the larger diameter bars could be due to the strength gradient through bar diameter during bending. During flexural tests, the outermost fibers are subject to the maximum stress. Thus, a smaller amount of fibers with respect to the total volume was under the maximum stress in the larger diameter bars. Moreover, the higher flexural-strength retention in the larger diameter bars can be correlated to the smaller reduction in the second moment of area of the conditioned specimens. Because only the bar surface was damaged, the fibers and matrix in the intact zone were undamaged and had the same initial mechanical properties, resulting in a better redistribution of load when the outer fibers progressively failed.

Fig. 8(c) shows an opposite trend than do Figs. 8(a and b), in which the smaller diameter bars exhibited higher tensile-strength retention. This figure shows that the #3 bars retained more than 95% of their tensile strength, while the #8 bars retained only 88%. This trend can be attributed to the shear lag effect. Archillides and Pilakoutas (2004) indicated that larger diameter FRP bars had more significant shear lag-effect than smaller diameter bars and had a noticeable effect on tensile strength. Castro and Carino (1998) further mentioned that the efficiency of the stress transfer from the bar surface to the interior fibers influences the mechanical properties of the FRP bars. Because of the shear lag effect, the outer fibers experience higher stresses than the inner fibers, even though all fibers resist the tensile load at the same time. Because of conditioning in alkaline solution, the outer surface of the bars was affected and might have decreased mechanical properties. As the specimens were loaded, the outer fibers, which were subjected to higher stress, initiated failure and the breakage moved instantly to the inner fibers, leading to the sudden and catastrophic failure of the GFRP bars. This also explains why the tensile strength was

Table 5. Estimated Affected Portion of the GFRP Bars

Affected portion	#3	#4	#5	#6	#8
Area ($\times 10^{-3}$ mm ²)	138.2	186.2	270.0	111.7	223.0
Thickness ($\times 10^{-3}$ mm)	4.4	4.7	5.4	1.9	2.8
Percentage of the nominal diameter	0.05	0.04	0.03	0.01	0.01

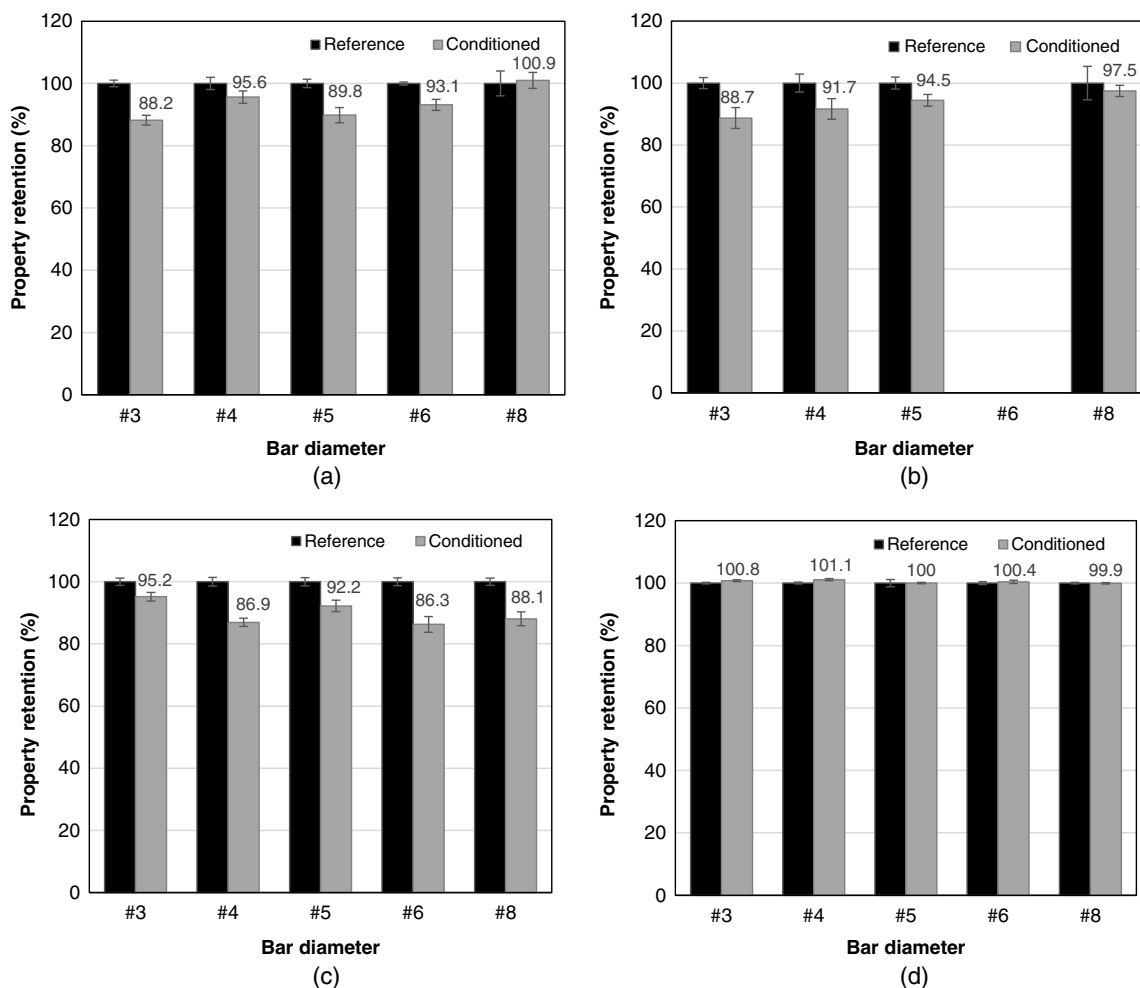


Fig. 8. Property retention of GFRP bars with different diameters: (a) interlaminar shear strength; (b) flexural strength; (c) tensile strength; (d) modulus of elasticity

more affected by the conditioning than were the interlaminar shear strength and flexural strength.

From the previous results, it can be concluded that the conditioning in alkaline solution affected the flexural, ILSS, and tensile strength properties of the GFRP bars, but not their moduli of elasticity. Moreover, the strength-retention limit was affected by bar size. The ILSS and flexural strength of the smaller diameter bars was affected more than the larger diameter ones. This is in contrast to tensile strength; the larger diameter bars had lower tensile strength retention. These results suggest that the conclusions drawn by most studies based on smaller diameter FRP bars do not apply to larger diameters. Based on the results in this study, it is more reasonable to use larger diameter bars in assessing the tensile-strength retention of FRP bars exposed to severe environmental conditions. Nevertheless, note should be taken of the very high load required to cause failure of larger bars. That notwithstanding, the strength retention in all the bar diameters considered in this study is significantly higher than the 0.70 environmental reduction coefficient required by several codes.

Scanning Electron Microscopy and Fourier Transformed Infrared Spectroscopy Observations

Scanning electron microscopy (SEM) observations were performed to assess the microstructure of the GFRP bars before and after

conditioning. All of the specimens observed under SEM were cut, polished, and coated with a thin layer of gold-palladium using a vapor-deposit process. Microstructural observations were performed on a JEOL JSM-840A SEM. Similarly, Fourier transformed infrared spectroscopy (FTIR) was conducted for each bar diameter to study the changes in the chemical composition of the matrix at the bar surface. These observations were conducted to determine the potential degradation of the polymer matrix, glass fibers, or interface, as applicable, due to the penetration of the alkaline solution. The aim was to link these observations to the possible evolution of mechanical properties and chemical composition of the bars after conditioning.

SEM

Figs. 9 and 10 show the SEM micrographs of the cross section of the reference and conditioned GFRP bars, respectively. As shown in Figs. 9(a and b), there were no pores observed in the center or near the surface of the reference bars. There was also no noticeable gap at the fiber-matrix interface, indicating excellent adhesion between the fibers and matrix. This is also true in the center of the conditioned GFRP bars [Fig. 10(a)]. Small gaps between the fibers and matrix were observed near the exposed surface in some conditioned specimens [Fig. 10(b)], suggesting that the bars were affected after exposure to the alkaline solution. This damage to the fiber-matrix interface reduces the transfer of loads between fibers

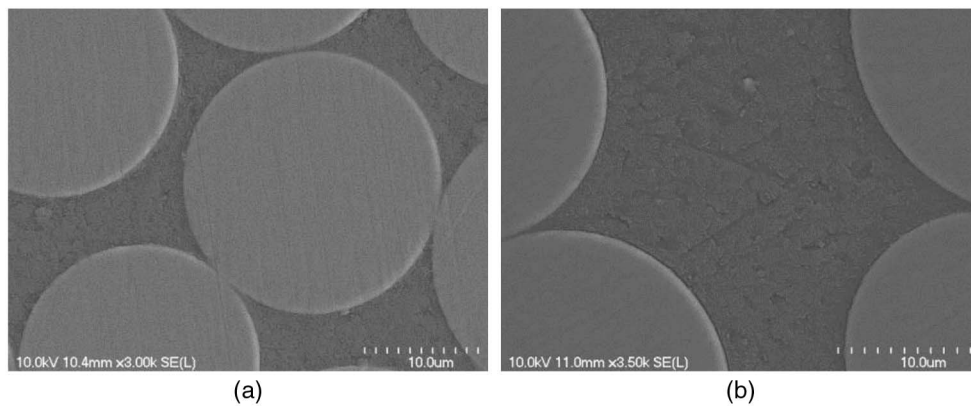


Fig. 9. SEM micrographs of the reference GFRP bars: (a) center of the bar; (b) near surface of the bar

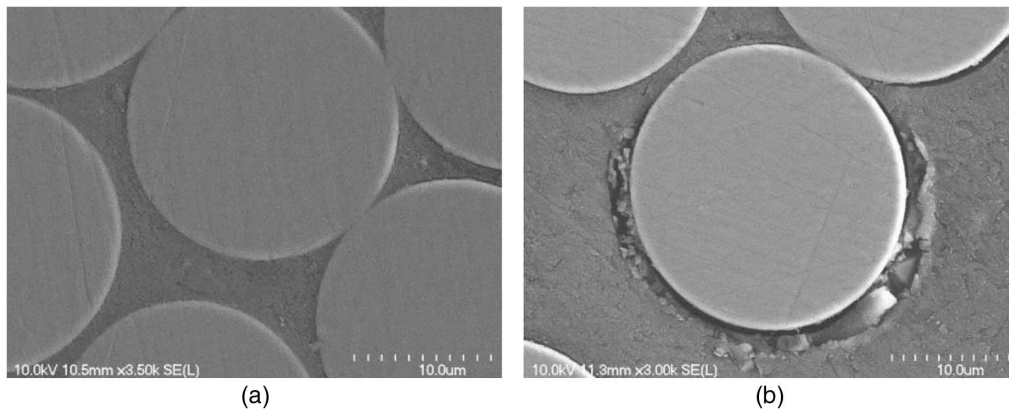


Fig. 10. SEM micrographs of the conditioned GFRP bars: (a) center of the bar; (b) near surface of the bar

and results in the reduction in strength properties. However, Benmokrane et al. (2002) indicated that the mechanical properties of GFRP bars are controlled by the fiber component. If the fibers have not deteriorated, FRP bars will preserve most of their mechanical strength and will be able to support loads. If the protective resin degrades, the bonding between the matrix and fibers

located on the outer part of the bar will gradually reduce and bar resistance will start to decrease. This is the most probable reason why the flexural, ILSS, and tensile strength properties of GFRP bars decreased after alkaline conditioning for three months. Nevertheless, the strength reduction is small as the SEM observation confirmed that only the exposed surface of the GFRP bars was affected,

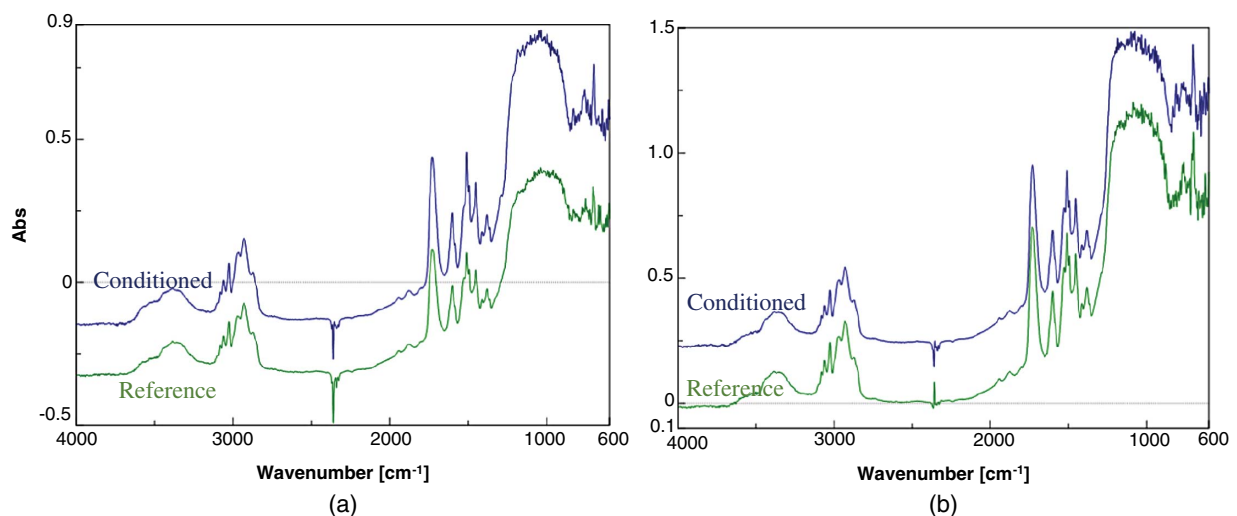


Fig. 11. FTIR spectrum of the GFRP bars before and after conditioning: (a) #3 GFRP bars; (b) #8 GFRP bars

not the core section. This exposed surface can be considered as being an all-resin surface, which does not contribute much in resisting the applied tensile load.

FTIR

Fig. 11 shows the infrared spectroscopy spectra of the surface of the reference and conditioned GFRP bars. Only the FTIR for the #3 and #8 bars are provided for comparison. The FTIR for both bar diameters shows no clear differences between the infrared spectra of the reference and conditioned GFRP bars. Moreover, the FTIR did not show any significant changes in the chemical structure after exposure to the alkaline solution. These observations indicate that the surface of the bars exposed to the alkaline solution for three months at 60°C had not been chemically modified and might possess the same initial mechanical properties. These results further confirm that the degradation of the matrix remains only at the exposed surface of the GFRP bars. This supports the findings by Nkurunziza et al. (2005) that vinylester epoxies are very resistant to chemical attack, which improves the environmental resistance of FRP bars made with these resins.

Conclusions

The effects of diameter on the physical, mechanical, and durability properties of sand-coated GFRP bars made of continuous boron-free glass fibers (EC-R) impregnated in a vinylester-based resin matrix were investigated. Based on the results of this study, the following conclusions were drawn for the tested GFRP bars:

- With the bar sizes considered, bar diameter did not affect fiber content, transverse coefficient of thermal expansion, porosity, or glass transition temperature. Conversely, the water absorption was found to decrease as the diameter increased. This can be correlated to the ratio of the surface area to the volume (shape ratio) of the GFRP bars;
- Tensile strength and modulus of the reference bars were not significantly affected by the cross-sectional size, but a size effect was observed for interlaminar shear strength and flexural strength. The consistency in the measured tensile properties for GFRP bars with different diameters is due to the efficient stress transfer from the bar surface to the center. Conversely, the higher probability of defects contained in the larger diameter bars may have caused the lower interlaminar shear strength and flexural strength in comparison to the smaller diameter bars;
- Elastic moduli of the reference and conditioned bars were same for all bar diameters, which is due to nearly the same fiber content of the GFRP bars;
- Interlaminar shear strength and flexural strength of the larger diameter GFRP bars were less affected after exposure to the alkaline solution than the smaller bar diameter. The higher interlaminar shear and flexural strength retention for the larger bar sizes was attributable to the lower affected thickness. As a result, the penetrated area was proportionally small relative to the total cross-sectional area of the bar;
- Tensile-strength retention was highest for the smallest diameter bar. This suggests that the impact of conditioning on the tensile properties of GFRP bars is expected to be greater for larger than smaller diameters; and
- The scanning electron microscope showed some small gaps between the fibers and matrix only at the exposed surface of the bars while the FTIR observations showed no changes in the chemical structure after conditioning in the alkaline solution for 90 days at 60°C. This shows that the degradation remained at the surface for all the bar diameters.

The variations in the physical and mechanical properties of the GFRP bars investigated in this study, from one diameter to another, remained low. Thus, the suggestions of the current standards and specifications to not relate the strength-retention limit to the size of the FRP bars are acceptable. However, further research is needed to investigate other bar types and diameters to clearly determine how the diameter might affect the design of GFRP-reinforced concrete structures.

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